# AP Calculus BC Key Points

Saijie Pan

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## Contents

# 1 Unit 1 Limit and Continuity

## 1.1 Definition of Limit

For a function $f(x)$, if as $x$ approaches $a$, $f(x)$ approaches $L$, i.e. $|f(x) - L|$ becomes arbitrarily small, then we say

$$\lim_{x \to a} f(x) = L$$

Equivalently,

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L$$

Note $\lim_{x \to a} f(x)$ can exist even if $f(a)$ is not defined.
For example, $\lim_{x \to 0} \frac{\sin x}{x} = 1$, $\lim_{x \to 0^+} \frac{|x|}{x} = 1$, and $\lim_{x \to 0^-} \frac{|x|}{x} = -1$.

## 1.2 When Limit Does Not Exist

- Oscillating: $\lim_{x \to 0} \sin \frac{1}{x}$
- Infinite: $\lim_{x \to 0} \frac{1}{x}$
- Left limit $\neq$ right limit: $\lim_{x \to 0} \frac{|x|}{x}$

## 1.3 Continuity at a Point

If $\lim_{x \to a} f(x) = f(a) = L$, then $f(x)$ is continuous at $x = a$.

## 1.4 Discontinuity (Three Types)

Three types of discontinuity:

- removable: limit exists but is discontinuous ($f(x)$ not defined or $f(a) \neq$ limit)
- jump: one-sided limits exist but left limit $\neq$ right limit
- infinite: at least a one-sided limit is infinite

## 1.5 Horizontal Asymptotes

If $\lim_{x \to \pm\infty} f(x) = L$, then there exists a horizontal asymptote $y = L$. There are at most two horizontal asymptotes for each function.

## 1.6 Vertical Asymptotes

If any one-sided limit $\lim_{x \to a^{\pm}} f(x) = \pm\infty$, then there exists a vertical asymptote $x = a$. For example, $y = \tan x$ has infinite number of vertical asymptotes.

## 1.7 Squeeze Theorem

If $a_n \leq b_n \leq c_n$ and $\lim_{x \to x_0} a_n = \lim_{x \to x_0} c_n = L$, then $\lim_{x \to x_0} b_n = L$.

For example, since $-\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$ for all positive $x$ and $\lim_{x \to \infty} \pm\frac{1}{x} = 0$, $\lim_{x \to \infty} \frac{\sin x}{x} = 0$

## 1.8 Intermediate Value Theorem (IVT)

- Condition is the most important: **continuous function**
- A function $f(x)$ that is continuous on $[a, b]$ takes on every $y$-value between $f(a)$ and $f(b)$.

# 2 Unit 2 Differentiation: Definition and Fundamental Properties

## 2.1 Average Rate of Change (AROC)

- Definition: $\text{AROC} = \dfrac{f(x_2) - f(x_1)}{x_2 - x_1}$
- Application: approximate $f'(a)$ for $x_1 \le a \le x_2$

## 2.2 Derivative: (Instantaneous) Rate of Change

$$\text{AROC} = \frac{\Delta y}{\Delta x}$$

$$\text{IROC} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$

## 2.3 Limit Definition of Derivative

The derivative of $f(x)$ at $x = a$ is

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

The derivative of $f(x)$ is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

## 2.4 Geometric Interpretation of Derivative

$f'(a)$: **slope** of the line tangent to $f(x)$ at $x = a$

Note the equation of tangent line is $y - f(a) = f'(a)(x - a)$ using point-slope form.

## 2.5 Connecting Differentiability and Continuity

differentiable at $x = a$ $\quad\Rightarrow\quad$ continuous at $x = a$ $\quad\Rightarrow\quad$ limit exists at $x = a$

## 2.6 Power Rule

$$\frac{d}{dx} x^n = n x^{n-1}$$

For example, $\left(\dfrac{1}{\sqrt{x}}\right)' = \left(x^{-\frac{1}{2}}\right)' = -\dfrac{1}{2} x^{-\frac{3}{2}}$

### 2.7 Important Derivative Formulas

$$\frac{d}{dx}\sin x = \cos x$$
$$\frac{d}{dx}\cos x = -\sin x$$
$$\frac{d}{dx}e^x = e^x$$
$$\frac{d}{dx}a^x = a^x \ln a$$
$$\frac{d}{dx}\ln x = \frac{1}{x}$$

### 2.8 Product Rule

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$

### 2.9 Quotient Rule

$$\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$$

### 2.10 Derivative of $\tan x$, $\cot x$, and $\sec x$

$$(\tan x)' = \left(\frac{\sin x}{\cos x}\right)' = \frac{1}{\cos^2 x} = \sec^2 x$$

$$(\cot x)' = \left(\frac{\cos x}{\sin x}\right)' = -\frac{1}{\sin^2 x} = -\csc^2 x$$

$$(\sec x)' = \left(\frac{1}{\cos x}\right)' = \frac{\sin x}{\cos^2 x} = \sec x \tan x$$

# 3 Unit 3 Differentiation: Composite, Implicit, and Inverse Functions

## 3.1 Chain Rule

The derivative of a composite function $y = f(g(x))$:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$[f(g(x))]' = f'(g(x))g'(x)$$

## 3.2 Implicit Differentiation

The chain rule is the basis. Three steps:

1. differentiate both sides of the equation with respect to $x$
2. collect terms containing $\frac{dy}{dx}$ on one side
3. find expression of $\frac{dy}{dx}$

For example,

$$x^2 + xy^2 + y^3 = 0$$

$$2x + 1 \cdot y^2 + x \cdot 2y \cdot \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

$$(2xy + 3y^2)\frac{dy}{dx} = -(2x + y^2) \Rightarrow \frac{dy}{dx} = -\frac{2x + y^2}{2xy + 3y^2}$$

## 3.3 Differentiating Inverse Function

For function $y = f(x)$ and inverse function $x = f^{-1}(y)$

$$\left(f^{-1}\right)'(y_0) = \left.\frac{dx}{dy}\right|_{y=y_0} = 1/\left.\frac{dy}{dx}\right|_{x=x_0} = 1/f'(x_0)$$

where $y_0 = f(x_0)$ and $x_0 = f^{-1}(y_0)$.

**Ex.** (2014BC28) The function $h(x) = x^5 + 3x - 2$ and $h(1) = 2$. If $h^{-1}$ is the inverse of $h$, what is the value of $(h^{-1})'(2)$?

Solution: $(h^{-1})'(2) = 1/h'(1) = 1/(5 + 3) = 1/8$

## 3.4 Derivative of Inverse Trig Functions

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1 - x^2}}$$

$$\frac{d}{dx} \arctan x = \frac{1}{1 + x^2}$$

### 3.5 Higher Order Derivatives

$f''(x)$ is the derivative of $f'(x)$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

$f^{(n)}(x)$ is the derivative of $f^{(n-1)}(x)$:

$$\frac{d^ny}{dx^n} = \frac{d}{dx}\left(\frac{d^{n-1}y}{dx^{n-1}}\right)$$

# 4 Unit 4 Contextual Applications of Differentiation

## 4.1 Straight-Line Motion

- Velocity is the derivative of position with respect to time: $v = \dfrac{dx}{dt}$
- Acceleration is the derivative of velocity with respect to time: $a = \dfrac{dv}{dt} = \dfrac{d^2x}{dt^2}$
- Scalar quantities: distance, speed
- Vector quantities: displacement, velocity

## 4.2 Related Rates

Connecting two rates (derivatives) with respect to time.

Solving related rates problems:

1. find related equations
2. implicit differentiation with respect to $t$ (refer to 3.2)
3. represent unknown with know derivatives

## 4.3 Linear Approximation

Equation of the line tangent to $f(x)$ at $(a, f(a))$:

$$y - f(a) = f'(a)(x - a)$$

$$y = f(a) + f'(a)(x - a)$$

Linear approximation of $f(x)$:

$$f(x) \approx f(a) + f'(a)(x - a)$$

Refer to 7.3 Euler's method and 10.6.2 Taylor polynomial.

## 4.4 L'Hospital's Rule

For "$\frac{0}{0}$" or "$\frac{\infty}{\infty}$" form, $\displaystyle\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$

- never forget to check conditions
- never forget limit notations

# 5 Unit 5 Analytical Applications of Differentiation

## 5.1 Mean Value Theorem (MVT)

Condition is important: $f(x)$ is **differentiable**
There exists $c$ where IROC equals AROC: $f'(c) = \dfrac{f(b) - f(a)}{b - a}$

## 5.2 Extreme Value Theorem (EVT)

EVT guarantees **both global/absolute extrema** (max and min) of a **continuous** function in **closed** interval.

## 5.3 Critical Points

Definition: $x$ coordinates where $f'(x) = 0$ or does not exist.

## 5.4 Increasing/Decreasing Intervals of Differentiable Function

- $f'(x) > 0 \Rightarrow f(x)$ is increasing
- $f'(x) < 0 \Rightarrow f(x)$ is decreasing

## 5.5 First Derivative Test to Determine Relative Extrema

Find **critical point** $x_c$ first!

- $f'(x)$ changes from negative to positive, $f(x)$ has a local min at $x = x_c$
- $f'(x)$ changes from positive to negative, $f(x)$ has a local max at $x = x_c$

## 5.6 Candidates Test to Determine Global Extrema

Global/absolute extrema only exists at critical points or endpoints

- Find all critical points and two endpoints
- List the values at critical points and endpoints
- max value of all candidates is global max while min value is global min

### 5.7 Determine Concavity

- **preferred**
  - $f'(x)$ is increasing $\Rightarrow$ the graph of $f(x)$ is concave up
  - $f'(x)$ is decreasing $\Rightarrow$ the graph of $f(x)$ is concave down
- If $f''(x)$ exists
  - $f''(x) > 0 \Rightarrow$ the graph of $f(x)$ is concave up
  - $f''(x) < 0 \Rightarrow$ the graph of $f(x)$ is concave down

### 5.8 Point of Inflection

- Definition: where concavity changes (i.e. $f'(x)$ changes from increasing/decreasing to decreasing/increasing)
- Note: if $f'' = 0$, there is not necessarily a point of inflection. For example $f(x) = x^4$ has $f''(0) = 0$, however there is no point of inflection at $x = 0$.

### 5.9 Second Derivative Test to Determine Local Extrema

- if $f''(a) > 0$ and $f'(a) = 0$ (critical point), $f(x)$ has a local min at $x = a$
- if $f''(a) < 0$ and $f'(a) = 0$ (critical point), $f(x)$ has a local max at $x = a$

Note if $f''(a) = 0$, the second derivative test fails and use first derivative test to find local extrema.

### 5.10 Optimization Problems

1. Understand the problem
2. Introduce notations to define variables
3. Formulate objective function
4. Reduce variables and keep only one
5. Use Calculus (refer to 5.6 determine global extrema)

# 6 Unit 6 Integration and Accumulation of Change

## 6.1 Calculate Definite Integral by Geometry

Definite Integral = signed area (area above X-axis − area below X-axis)

Regular shapes: circle, semi-circle, rectangle, triangle, trapezoid$\cdots$

## 6.2 Properties of Definite Integral

- Integral of a constant

$$\int_a^b C\,dx = C(b-a)$$

- Linearity

$$\int_a^b (f(x)+g(x))\ dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

$$\int_a^b C\cdot f(x)\,dx = C\cdot \int_a^b f(x)\,dx$$

- Addition of adjacent intervals

$$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$$

- Reversing limits

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

## 6.3 Four Types of Riemann Sum

- left Riemann Sum: $\displaystyle\sum_{i=1}^{n} f(x_{\text{left}})\cdot \Delta x_i$
- right Riemann Sum: $\displaystyle\sum_{i=1}^{n} f(x_{\text{right}})\cdot \Delta x_i$
- midpoint Riemann Sum: $\displaystyle\sum_{i=1}^{n} f(\frac{x_{\text{left}}+x_{\text{right}}}{2})\cdot \Delta x_i$
- trapezoidal Riemann Sum: $\displaystyle\sum_{i=1}^{n} \frac{f(x_{\text{left}})+f(x_{\text{right}})}{2}\cdot \Delta x_i$

Riemann sum $\approx \displaystyle\int_a^b f(x)\,dx$. Underestimate (sum < integral) or Overestimate (sum > integral)?

- If $f(x)$ is increasing, the right (left) Riemann sum is an overestimate (underestimate).
- If $f(x)$ is decreasing, the left (right) Riemann sum is an overestimate (underestimate).
- If $f(x)$ is concave up (down), the trapezoidal sum is an overestimate (underestimate).

### 6.4 Fundamental Theorem of Calculus Part 1(FTC1)

$$\frac{d}{dx}\left(\int_a^x f(t)\,dt\right) = f(x)$$

Applying chain rule,

$$\frac{d}{dx}\left(\int_a^{g(x)} f(t)\,dt\right) = f(g(x))\cdot g'(x)$$

Example: $f(x) = \int_1^{\sin x}(1+t)^3\,dt$, find $f'(\pi)$.
Solution: $f'(x) = (1+\sin x)^3\cdot\cos x,\ f'(\pi) = (1+0)^3\cdot(-1)$

### 6.5 Fundamental Theorem of Calculus Part 2 (FTC2)

$$\int_a^b f'(x)\,dx = f(x)\Big|_a^b = f(b) - f(a)$$

### 6.6 Important Integral Formulas

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C, \text{ for } n \neq -1 \qquad \int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^x\,dx = e^x + C \qquad \int a^x\,dx = \frac{a^x}{\ln a} + C$$

$$\int \sin x\,dx = -\cos x + C \qquad \int \cos x\,dx = \sin x + C$$

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + C \qquad \int \frac{1}{1+x^2}\,dx = \arctan x + C$$

### 6.7 Integral Methods

- U-Substitution $\int x^2(x^3+1)^5\,dx$

$$u = x^3+1 \Rightarrow du = 3x^2\,dx$$

$$\int x^2(x^3+1)^5\,dx = \int u^5\,\frac{du}{3} = \frac{(x^3+1)^6}{18} + C$$

- Long Division $\int \frac{x^2+3x+1}{x+1}\,dx$

$$\int \frac{x^2+3x+1}{x+1}\,dx = \int \frac{(x+1)(x+2)-2+1}{x+1}\,dx = \int (x+2)\,dx - \int \frac{1}{x+1}\,dx$$

- Completing Squares $\int \frac{1}{x^2+2x+2}\,dx$

$$\int \frac{1}{x^2+2x+2}\,dx = \int \frac{1}{(x+1)^2+1}\,dx$$

Using U-Substitution, $\int \frac{1}{(x+1)^2+1}\,dx = \arctan(x+1) + C$

- Partial Fraction $\int \frac{2x+1}{x^2+x}\,dx$

$$\frac{2x+1}{x^2+x} = \frac{2x+1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1} = \frac{1}{x} + \frac{1}{x+1}$$

$$\int \frac{2x+1}{x^2+x}\,dx = \int \left(\frac{1}{x} + \frac{1}{x+1}\right)\,dx = \ln|x| + \ln|x+1| + C$$

- Integration By Parts $\int uv'\,dx = uv - \int vu'\,dx$

  Example 1: $\displaystyle\int x\cos x\,dx$

$$u = x, v' = \cos x \qquad \Rightarrow \qquad u' = 1, v = \sin x$$

$$\int x\cos x\,dx = x\cdot\sin x - \int \sin x\cdot 1\,dx = x\cdot\sin x + \cos x + C$$

  Example 2: $\displaystyle\int \arcsin x\,dx$

$$u = \arcsin x, v' = 1 \qquad \Rightarrow \qquad u' = \frac{1}{\sqrt{1-x^2}}, v = x$$

$$\int \arcsin x\,dx = x\cdot\arcsin x - \int \frac{x}{\sqrt{1-x^2}}\,dx$$

  using U-substitution to solve $\int \frac{x}{\sqrt{1-x^2}}\,dx = -\sqrt{1-x^2} + C$

## 6.8 Improper Integral

Convert to limit format first!

$$\int_0^\infty e^{-x}\,dx = \lim_{b\to\infty}\int_0^b e^{-x}\,dx = \lim_{b\to\infty}(-e^{-x})\Big|_0^b = \lim_{b\to\infty}(-e^{-b}+1) = 1$$

$$\int_0^1 \frac{1}{\sqrt{x}}\,dx = \lim_{a\to 0^+}\int_a^1 \frac{1}{\sqrt{x}}\,dx = \lim_{a\to 0^+}(2\sqrt{x})\Big|_a^1 = \lim_{a\to 0^+}(2-2\sqrt{a}) = 2$$

Useful conclusions:

$$\int_1^\infty \frac{1}{x^p}\,dx = \begin{cases} \frac{1}{p-1} & \text{if } p > 1 \\ \text{diverges} & \text{if } p \le 1 \end{cases}$$

Note: An improper integral is said to **converge** if the limit of the integral exists. An improper integral is said to **diverge** when the limit of the integral fails to exist.

# 7 Unit 7 Differential Equation

## 7.1 Slope Field

Consider special points (axis, $y = \pm x$) and special slopes (0) first.

## 7.2 Solving Differential Equation

1. separation of variables ($y$ terms on the left side and $x$ terms on the right side)
2. add integral symbol $\int$ on both sides
3. calculate anti-derivatives on both sides and add constant $C$ on the right side
4. plug in initial condition and calculate $C$
5. find the expression of $y = f(x)$

## 7.3 Euler's Method

Linear approximation step by step:

- start from initial condition $(x_0, f(x_0))$
- $f(x_1) \approx f(x_0) + f'(x_0)(x_1 - x_0)$
- $f(x_2) \approx f(x_1) + f'(x_1)(x_2 - x_1)$
- $\cdots$

For example, given $f(0)$, approximate $f(1)$ using 4 steps. The step is $\frac{1-0}{4} = 0.25$. We need to calculate $f(0.25)$, $f(0.5)$, $f(0.75)$, and $f(1)$. $f'(x)$ is usually given in the form of differential equation.

## 7.4 Logistic Equation

- Remember the form of logistic equation $\dfrac{dP}{dt} = kP(M - P)$ where $M$ is carrying capacity.
- Know that the graph of $P(t)$ is an S-shaped curve and there is a point of inflection when $P = \frac{M}{2}$, at which $\frac{dP}{dt}$ reaches max.
- Examples:
  $$\frac{dy}{dx} = y(1 - \frac{y}{100})$$
  $$\frac{dx}{dt} = x(100 - x)$$
  Both equations above are logistic equations with carrying capacity $M = 100$.
  $$\frac{dy}{dt} = 3t(10 - t)$$
  The equation above is not a logistic equation.

# 8 Unit 8 Applications of Integration

## 8.1 Average Value of a Function on an Interval

$$\overline{f(x)} = \frac{\int_a^b f(x)\,dx}{b-a}$$

Don't confuse it with average rate of change (see 2.1).

## 8.2 Position, Velocity, and Acceleration

Define position as $x(t)$, velocity as $v(t) = x'(t)$, and acceleration as $a(t) = v'(t) = x''(t)$.

$$x(t_2) - x(t_1) = \int_{t_1}^{t_2} v(t)\,dt \qquad \Rightarrow \qquad x(t_2) = x(t_1) + \int_{t_1}^{t_2} v(t)\,dt$$

$$v(t_2) - v(t_1) = \int_{t_1}^{t_2} a(t)\,dt \qquad \Rightarrow \qquad v(t_2) = v(t_1) + \int_{t_1}^{t_2} a(t)\,dt$$

- **displacement** (integral of velocity): $\int_{t_1}^{t_2} v(t)\,dt$
- **distance** traveled (integral of **speed** $|v|$): $\int_{t_1}^{t_2} |v(t)|\,dt$

## 8.3 Area Between Curves Expressed as Functions

The area of the region bounded above by $U(x)$, below by $D(x)$, left by $x = x_1$, and right by $x = x_2$:

$$S = \int_{x_1}^{x_2} (U(x) - D(x))\,dx$$

The area of the region bounded above by $y = y_2$, below by $y = y_1$, left by $L(y)$, and right by $R(y)$:

$$S = \int_{y_1}^{y_2} (R(y) - L(y))\,dy$$

## 8.4 Volume with Cross Sections

- Determine $dx$ or $dy$: cross section is perpendicular to which ($x$ or $y$) axis
- Find cross section area expression $A(x)$ or $A(y)$
- The volume $V = \displaystyle\int_{a_1}^{a_2} A(x)\,dx$ or $\displaystyle\int_{b_1}^{b_2} A(y)\,dy$

A list of cross sections: squares, rectangles, triangles, and semicircles.

## 8.5 Volume with Disc Method: Revolving around Any Axes

- Determine $dx$ or $dy$ from axis of revolution (i.e. $dx$ if about $x$-axis or horizontal line $y = a$; $dy$ if about $y$-axis or vertical line $x = b$)
- Find radius of disc $R$, which is the distance between the curve and axis of revolution
- The area of the cross section $A = \pi R^2$
- The volume $V = \displaystyle\int_{a_1}^{a_2} \pi R^2 \, dx$ or $\displaystyle\int_{b_1}^{b_2} \pi R^2 \, dy$

## 8.6 Volume with Washer Method: Revolving around Any Axes

Similar to disc method, actually disc method is a special case of washer method

- Determine $dx$ or $dy$ from axis of revolution (i.e. $dx$ if about $x$-axis or horizontal line $y = a$; $dy$ if about $y$-axis or vertial line $x = b$)
- Find inner radius $r$, which is the distance between the inner curve and axis of revolution, and outer radius $R$, which is the distance between the outer curve and axis of revolution
- The area of the cross section $A = \pi(R^2 - r^2)$
- The volume $V = \displaystyle\int_{a_1}^{a_2} \pi(R^2 - r^2) \, dx$ or $\displaystyle\int_{b_1}^{b_2} \pi(R^2 - r^2) \, dy$

## 8.7 The Arc Length of a Curve or Distance Travelled

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \cdot dx$$

So the arc length is

$$\int ds = \int \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

# 9 Unit 9 Parametric Equation and Polar Curve

## 9.1 Defining Parametric Equations

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases}$$

## 9.2 First Derivative of Parametric Equations

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{y'(t)}{x'(t)}$$

## 9.3 Second Derivative of Parametric Equations

1. Find first derivative $\frac{dy}{dx} = f(t)$, refer to 9.2
2. Second derivative is the derivative of the first derivative
$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{df}{dx} = \frac{f'(t)}{x'(t)}$$

## 9.4 Arc Lengths of Curves Given by Parametric Equations

- refer to arc length formulas in 8.7
- Formula in parametric equations

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \cdot dt$$

$$\int ds = \int_{t_1}^{t_2} \sqrt{(x'(t))^2 + (y'(t))^2}\, dt$$

## 9.5 Motion Problems Using Parametric and Vector-Valued Functions

- position $\langle x(t), y(t)\rangle$
- velocity $\langle x'(t), y'(t)\rangle = \langle v_x(t), v_y(t)\rangle$
- acceleration $\langle x''(t), y''(t)\rangle$
- speed $|v| = \sqrt{(x'(t))^2 + (y'(t))^2}$
- distance travelled $\int |v|\, dt = \int_{t_1}^{t_2} \sqrt{(x'(t))^2 + (y'(t))^2}\, dt$
- final position $x(t_2) = x(t_1) + \int_{t_1}^{t_2} x'(t)\, dt$ and $y(t_2) = y(t_1) + \int_{t_1}^{t_2} y'(t)\, dt$

### 9.6 Polar Coordinate

- Definition $(r, \theta)$
- Convert to parametric equations

$$\begin{cases} x(\theta) = r\cos\theta \\ y(\theta) = r\sin\theta \end{cases}$$

- Area bounded by polar curves $r = f(\theta)$

$$S = \frac{1}{2}\int_{\theta_1}^{\theta_2} r^2\,d\theta = \frac{1}{2}\int_{\theta_1}^{\theta_2} f^2(\theta)\,d\theta$$

- Differentiating in polar form (slope of tangent line)
  1. convert polar curve $r = f(\theta)$ to parametric equations

$$\begin{cases} x(\theta) = f(\theta)\cos\theta \\ y(\theta) = f(\theta)\sin\theta \end{cases}$$

  2. find the first derivative of the above parametric equations (refer to 9.2)

$$\frac{dy}{dx} = \frac{y'(\theta)}{x'(\theta)} = \frac{(f(\theta)\sin\theta)'}{(f(\theta)\cos\theta)'}$$

# 10 Unit 10 Infinite Series

## 10.1 Test for Convergence/Divergence

### 10.1.1 Divergence Test (nth Term Test)

$$\lim_{n\to\infty} a_n \neq 0 \Rightarrow \sum a_n \text{ diverges}$$

Divergence test can **NOT** determine convergence.

### 10.1.2 Geometric Series

$$\sum ar^n = a + ar + ar^2 + \cdots$$

- If common ratio $|r| < 1$, then $\sum ar^n$ converges and $\sum ar^n = \dfrac{a}{1-r}$.
- If common ratio $|r| \geq 1$, then $\sum ar^n$ diverges.

### 10.1.3 p-Series

$$\sum \frac{1}{n^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots$$

- If $p = 1$, $\sum \dfrac{1}{n} = 1 + \dfrac{1}{2} + \dfrac{1}{3} + \cdots$ is also called harmonic series. Harmonic series diverges.
- If $p > 1$, p-series converges. For example, $\sum \dfrac{1}{\sqrt{n^3}}$ converges ($p = \frac{3}{2} > 1$).
- If $p \leq 1$, p-series diverges. For example, $\sum \dfrac{1}{\sqrt{n}}$ diverges ($p = \frac{1}{2} \leq 1$).

### 10.1.4 Integral Test

- Conditions: $a_n = f(n)$, the function $f$ is positive, continuous, and decreasing
- p-series test is proved by integral test.
- For example,

$$\int_2^\infty \frac{1}{x \ln x}\, dx \text{ diverges} \qquad \Rightarrow \qquad \sum_{n=2}^{\infty} \frac{1}{n \ln n} \text{ diverges}$$

$$\int_2^\infty \frac{1}{x(\ln x)^2}\, dx \text{ converges} \qquad \Rightarrow \qquad \sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2} \text{ converges}$$

### 10.1.5 Comparison Test

If $0 \leq a_n \leq b_n$, then

$$\sum b_n \text{ converges} \quad \Rightarrow \quad \sum a_n \text{ converges}$$
$$\sum a_n \text{ diverges} \quad \Rightarrow \quad \sum b_n \text{ diverges}$$

### 10.1.6 Limit Comparison Test

- For positive series, if $\lim\limits_{n\to\infty} \dfrac{a_n}{b_n} = L > 0$, then both series converge or both diverge.
- Use with "messy" algebraic series, usually compared to a p-series or a geometric series.

### 10.1.7 Alternating Series Test (Leibniz Test)

If $\sum(-1)^n a_n$ meets the following three conditions:

1. $a_n > 0$, which means $\sum(-1)^n a_n$ is an alternating series
2. $a_n$ is decreasing
3. $\lim\limits_{n\to\infty} a_n = 0$,

then the alternating series $\sum(-1)^n a_n$ converges.

### 10.1.8 Absolute Convergence Implies Convergence

$$\sum |a_n| \text{ converges} \quad \Rightarrow \quad \sum a_n \text{ converges}$$

### 10.1.9 Ratio Test

Works well for $a_n$ containing factorials and exponentials

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1 \quad \Rightarrow \quad \sum a_n \text{ converges}$$
$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1 \quad \Rightarrow \quad \sum a_n \text{ diverges}$$

## 10.2 Absolute/Conditional Convergence

Definition:

- If $\sum |a_n|$ converges, then $\sum a_n$ absolutely converges. (e.g. $\sum(-1)^n \dfrac{1}{n^2}$)

- If $\sum |a_n|$ diverges but $\sum a_n$ converges, then $\sum a_n$ conditionally converges. (e.g. $\sum (-1)^n \frac{1}{\sqrt{n}}$)

## 10.3 Power Series

### 10.3.1 Definition of Power Series

Definition: $\sum_{n=0}^{\infty} c_n (x-a)^n$, centered at $x = a$.

### 10.3.2 Radius of Convergence (ROC)

- When $|x-a| < R$, power series converges **absolutely** and when $|x-a| > R$, power series diverges. $R$ is defined as radius of converges.
- Use ratio test (refer to 10.1.9) to determine ROC

### 10.3.3 Interval of Convergence (IOC)

- determine radius of convergence $R$, so $|x-a| < R$
- check two endpoints ($x_1 = a - R$ and $x_2 = a + R$)

## 10.4 Taylor and Maclaurin Series

### 10.4.1 Definition of Taylor Series

Taylor series of $f(x)$ at $x = a$:

$$\begin{aligned} T(x) &= \sum_{n=0}^{\infty} c_n (x-a)^n \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \\ &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots \end{aligned}$$

$T(x)$ is a power series with coefficient $c_n = \frac{f^{(n)}(a)}{n!}$. If $c_n$ is given, we can easily calculate $f^{(n)}(a)$.

### 10.4.2 Definition of Maclaurin Series

Maclaurin series is a special case of Taylor series when $a = 0$

$$T(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots$$

### 10.4.3 Important Maclaurin Series

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

### 10.4.4 Geometric (Power) Series

Revisit geometric series

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots = \sum_{n=0}^{\infty} x^n \text{ for } |x| < 1$$

Note the interval of convergence $|x| < 1$ is very important.

## 10.5 Shortcuts to Finding Taylor Series

- multiplying

$$xe^x = x \sum_{n=0}^{\infty} \frac{x^n}{n!} = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n!}$$

- substitution

$$\frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty} (-x)^n = \sum_{n=0}^{\infty} (-1)^n x^n = 1 - x + x^2 - x^3 + \cdots$$

  Note it is valid only for $|-x| < 1$ (i.e. $|x| < 1$).

- integration

$$\ln(1+x) = \int \frac{1}{1+x} dx = \int \sum_{n=0}^{\infty} (-1)^n x^n \, dx = \sum_{n=0}^{\infty} \int (-1)^n x^n \, dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{(n+1)}$$

  Note the integration constant is zero since $\ln(1+0) = 0$.

- differentiation

$$\frac{1}{(1+x)^2} = -\left(\frac{1}{1+x}\right)' = -\left(\sum_{n=0}^{\infty} (-1)^n x^n\right)' = -\sum_{n=0}^{\infty} (-1)^n n x^{n-1} = 1 - 2x + 3x^2 + \cdots$$

## 10.6 Error Bound

### 10.6.1 Alternating Series Error

Applying alternating series test (3 conditions), $\sum a_n$ converges to $S$. Use $S_n = a_1 + a_2 + \cdots + a_n$ to approximate $S$, then the error

$$|S_n - S| < |a_{n+1}| \text{ (the next term)}$$

### 10.6.2 nth Degree Taylor Polynomial

$$T_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Note that nth degree Taylor polynomial uses all the Taylor series terms up to the term including the nth derivative.

### 10.6.3 Lagrange Error

Using $T_n(x)$ to approximate $f(x)$, the error

$$|T_n(x) - f(x)| \leq \frac{\max\left|f^{(n+1)}(z)\right|}{(n+1)!}|x-a|^{n+1}$$

Note that $z$ is between center $a$ and $x$, and $\max\left|f^{(n+1)}(z)\right|$ is usually given or easy to find.